

ELEC2302/9302 Signals and Systems — Quiz 2 practice questions, weeks 4 to 7¹

How to use this question bank: *don't* work through the questions from beginning to end. Start by doing one question from each group, identify the areas where you are less confident, and work on those first.

1. Fourier series (week 4)**Fundamental frequency of a sum of sinusoids**

1. A signal is given by

$$u(t) = \cos\left(\frac{2}{3}t + \frac{\pi}{6}\right) + \sin\left(\frac{4}{5}t + \frac{\pi}{4}\right).$$

Is $u(t)$ periodic, and if so what is its fundamental frequency ω_0 ?

- (a) Periodic, $\omega_0 = 1/15$ rad/s
- (b) Periodic, $\omega_0 = 4/5$ rad/s (the larger of the two frequencies)
- (c) Not periodic, because $2/3$ and $4/5$ are not integers
- (d) Periodic, $\omega_0 = 2/15$ rad/s

2. A signal is given by

$$u(t) = 2 + 7 \cos\left(\frac{1}{2}t + \theta_1\right) + 3 \cos\left(\frac{2}{3}t + \theta_2\right) + 5 \cos\left(\frac{7}{6}t + \theta_3\right).$$

Find its fundamental frequency ω_0 , in rad/s.

Answer: _____

3. A signal is given by $u(t) = \cos(3t) + \sin(\sqrt{5}t)$. Is $u(t)$ periodic?

- (a) Yes, periodic with $\omega_0 = 3$ rad/s, the smaller of the two frequencies
- (b) Yes – any sum of two sinusoids is periodic
- (c) Yes, periodic with $\omega_0 = \sqrt{5} - 3$ rad/s
- (d) No – a sum of two sinusoids is periodic only if the ratio of their frequencies is rational, and $3/\sqrt{5}$ is irrational

Fourier series coefficients

4. A periodic signal $u(t)$ of period T satisfies $u(t - T/2) = -u(t)$. Which of its Fourier coefficients c_k must be zero?

- (a) Only c_0
- (b) c_k for every even k (including c_0)
- (c) c_k for every odd k
- (d) None – the condition only affects the phase of c_k , not which ones vanish

¹Some questions are adapted from B. P. Lathi and R. A. Green, *Linear Systems and Signals*, 3rd ed. (chapters 2, 4, 6 and 7), from ELEC3004 *Signals, Systems and Control* at the University of Queensland (2013 and 2017), and from S. Boyd, EE102 *Introduction to Signals and Systems*, Stanford University.

5. The sawtooth wave $u(t) = 2At/T$ for $|t| < T/2$ (period T) has which of the following properties?
- (a) $b_k = 0$ for all k ; every c_k is purely real
 - (b) $a_k = 0$ for all k and $c_0 = 0$; every c_k is purely imaginary
 - (c) Only the odd-indexed c_k are nonzero, as for a square wave satisfying $u(t - T/2) = -u(t)$
 - (d) $c_0 = A$, the peak value of the ramp

6. A periodic signal is $u(t) = 1$ for $|t| < \pi/2$ and 0 for $\pi/2 < |t| < \pi$, repeated with period 2π : an even square pulse train with 50% duty cycle. Once its DC component is removed, which coefficients remain nonzero?
- (a) Only the odd-indexed b_k
 - (b) All a_k , $k \geq 1$, decaying as $1/k$
 - (c) Only $a_2, a_4, a_6, \dots$
 - (d) Only the odd-indexed a_k – the signal is even, and once the DC offset is removed it satisfies $u(t - T/2) = -u(t)$, which kills every even harmonic

7. An even, bipolar square wave ($+A$ for $|t| < T/4$, $-A$ for $T/4 < |t| < T/2$) has amplitude $A = 6$. Find c_5 , to 4 significant figures.

Answer: _____

8. A full-wave rectified sine $u(t) = A|\sin(\omega_m t)|$ has amplitude $A = 7$. Find c_3 , to 4 significant figures.

Answer: _____

9. A pulse train $u(t) = Ap_\tau(t)$ (period T , duty cycle $d = \tau/T = 1/3$) has Fourier coefficients $c_k = Ad \operatorname{sinc}(kd)$. For which smallest positive integer k does $c_k = 0$?

- (a) $k = 2$
- (b) c_k is never zero when $d = 1/3$
- (c) $k = 3$
- (d) $k = 1$

10. A periodic signal $u(t)$ with period T equals $+8$ for a quarter of each period and -2 for the remaining three-quarters. Find c_0 .

Answer: _____

11. A signal $u(t)$ has period $T = 2$ and is defined on one period by $u(t) = 1$ for $0 < t < 1$ and $u(t) = -1$ for $1 < t < 2$ (repeating). Its complex Fourier coefficient c_0 is:

- (a) $c_0 = -1$
- (b) $c_0 = 0.5$
- (c) $c_0 = 0$
- (d) $c_0 = 1$

12. A signal of period 1 equals $1/\sqrt{d}$ on $0 \leq t < d$ and 0 on the rest of each period, where $0 < d < 1$. For $d = 0.01$, find c_0 .
- (a) 0.01
 - (b) 0.1
 - (c) 10
 - (d) 1
13. In the exponential Fourier series $u(t) = \sum_k c_k e^{jk\omega_0 t}$ of a real signal, the term with index $k = -3$ is best interpreted as:
- (a) The complex conjugate of c_2
 - (b) A genuinely negative-frequency oscillation, physically distinct from the $k = +3$ component
 - (c) One half of the real sinusoidal component at frequency $3\omega_0$, which pairs with the $k = +3$ term to form $2|c_3| \cos(3\omega_0 t + \arg c_3)$
 - (d) A numerical artefact that can be discarded without changing $u(t)$
14. A periodic signal's complex Fourier coefficient is $c_1 = 2 - 3j$. Using the convention ($a_k = 2 \operatorname{Re} c_k$, $b_k = -2 \operatorname{Im} c_k$), the pair (a_1, b_1) is:
- (a) $(-6, 4)$
 - (b) $(2, -3)$
 - (c) $(4, 6)$
 - (d) $(4, -6)$
15. The sine-cosine Fourier series of a periodic signal $u(t)$ contains the term $3 \cos(2\omega_0 t) + 4 \sin(2\omega_0 t)$ (i.e. $a_2 = 3$, $b_2 = 4$). Find $|c_2|$.

Answer: _____

16. A periodic signal is

$$u(t) = 2 + 6 \cos(300t) + 8 \sin(500t).$$

Which set of Fourier coefficients describes it?

- (a) $\omega_0 = 100 \text{ rad/s}$, $c_0 = 2$, $c_1 = c_{-1} = 3$, $c_2 = -4j$, $c_{-2} = 4j$, all others zero
- (b) $\omega_0 = 100 \text{ rad/s}$, $c_0 = 2$, $c_3 = c_{-3} = 3$, $c_5 = 4j$, $c_{-5} = -4j$, all others zero
- (c) $\omega_0 = 100 \text{ rad/s}$, $c_0 = 2$, $c_3 = c_{-3} = 3$, $c_5 = -4j$, $c_{-5} = 4j$, all others zero
- (d) $\omega_0 = 100 \text{ rad/s}$, $c_0 = 2$, $c_3 = c_{-3} = 6$, $c_5 = -8j$, $c_{-5} = 8j$, all others zero

Parseval's theorem for periodic signals

17. Parseval's theorem for a periodic signal $u(t)$ states that its average power is:

- (a) $\int_{-\infty}^{\infty} |c_k| d\omega$
- (b) $|c_0|^2$ alone

$$(c) \sum_{k=-\infty}^{\infty} c_k$$

$$(d) \sum_{k=-\infty}^{\infty} |c_k|^2$$

18. An odd, bipolar square wave of amplitude $A = 1$ (period T , so $|u(t)| = 1$ always) has $c_k = -2j/(\pi k)$ for odd k , zero otherwise. What percentage of its total average power is carried by the fundamental component alone ($k = \pm 1$)? Give your answer to 4 significant figures.

Answer: _____

19. A rectangular pulse train $A p_\tau(t)$ of period T has duty cycle $d = \tau/T = 0.25$. What fraction of its total average power is carried by the DC term c_0 alone?

Answer: _____

Signals as vectors, projection and orthogonality

20. A square wave $u(t)$ has period $T = 2\pi$, with $u(t) = +1$ for $0 < t < \pi$ and $u(t) = -1$ for $\pi < t < 2\pi$. It is approximated by a single complex sinusoid at the fundamental frequency, $u(t) \approx c e^{jt}$, with the complex constant c chosen to minimise $\|u - c e^{jt}\|_P^2$. Find c .

Answer: _____

21. A real T -periodic signal $u(t)$ is approximated by $c \phi(t)$, with the real constant c chosen to minimise $\|u - c \phi\|_P^2$. At that optimal c , the error signal $e(t) = u(t) - c \phi(t)$ satisfies:

- (a) $e(t) = 0$ for every t
- (b) $\langle e, \phi \rangle_P = 0$: the error is orthogonal to ϕ
- (c) $\|e\|_P = \|u\|_P$
- (d) $\langle e, u \rangle_P = 0$: the error is orthogonal to u

22. Two real T -periodic signals $u(t)$ and $y(t)$ are orthogonal, $\langle u, y \rangle_P = 0$. Let $z(t) = u(t) + y(t)$. The average power $P(z)$ equals:

- (a) $P(u) - P(y)$
- (b) $P(u) P(y)$
- (c) $P(u) + P(y)$
- (d) $\left(\sqrt{P(u)} + \sqrt{P(y)}\right)^2$

2. Fourier transform (week 5)

Existence of the transform, and symmetry for real, even and odd signals

23. Which of the following signals does **not** possess a Fourier transform, not even one containing impulses?

- (a) $\cos(5t)$
- (b) $H(t)$
- (c) 1

(d) $e^{2t}H(t)$

24. $u(t)$ is real and odd, i.e. $u(-t) = -u(t)$. Which statement about $U(j\omega)$ must be true?

- (a) $U(j\omega)$ is real and an odd function of ω
- (b) $U(j\omega)$ is purely imaginary and an odd function of ω
- (c) $U(j\omega)$ is purely imaginary and an even function of ω
- (d) $U(j\omega)$ is real and an even function of ω

25. $u(t)$ is real. Which statement about its spectrum is always true?

- (a) Both $|U(j\omega)|$ and $\arg U(j\omega)$ are odd
- (b) Both $|U(j\omega)|$ and $\arg U(j\omega)$ are even
- (c) $|U(j\omega)|$ is even and $\arg U(j\omega)$ is odd
- (d) $|U(j\omega)|$ is odd and $\arg U(j\omega)$ is even

Area relations between u and $U(j\omega)$

26. $u(t) = 3e^{-2t}H(t) - 2e^{-5t}H(t)$. Find $U(0)$.

Answer: _____

27. $U(j\omega) = 4$ for $|\omega| < 1$ and 0 otherwise (i.e. $U(j\omega) = 4p_2(\omega)$). Find $u(0)$.

Answer: _____

28. Evaluate $\int_{-\infty}^{\infty} \text{sinc}(t) dt$.

Answer: _____

Transforms of standard signals

29. What is the Fourier transform of $u(t) = \delta(t - 5)$?

- (a) $e^{-j5\omega}$
- (b) $5e^{-j\omega}$
- (c) $\delta(\omega - 5)$
- (d) $e^{j5\omega}$

30. What is the Fourier transform of the constant signal $u(t) = 1$, and why is the weight 2π (not 1) attached to the impulse?

- (a) $U(j\omega) = 2\pi\delta(\omega)$, because $u(t)$ is synthesised from $u(t) = \frac{1}{2\pi} \int U(j\omega)e^{j\omega t} d\omega$, so a single exponential of amplitude 1 needs an impulse of area 2π
- (b) $U(j\omega) = \delta(\omega)$, because $u(t) = 1 = e^{j0 \cdot t}$ needs an impulse of unit strength
- (c) $U(j\omega) = \delta(\omega)/2\pi$
- (d) $U(j\omega) = 2\pi\delta(\omega)$, because $\int_{-\infty}^{\infty} 1 dt = 2\pi$

31. What is the Fourier transform of $u(t) = \text{sinc}(t)$?

- (a) $U(j\omega) = \text{sinc}(\omega)$
- (b) $U(j\omega) = p_\pi(\omega)$
- (c) $U(j\omega) = p_{2\pi}(\omega)$, i.e. 1 for $|\omega| < \pi$ and 0 otherwise
- (d) $U(j\omega) = \pi p_{2\pi}(\omega)$

32. $u(t) = p_6(t)$ (a pulse of width 6, centred at the origin). What is $U(j\omega)$?

- (a) $6 \text{sinc}\left(\frac{\omega}{2\pi}\right)$
- (b) $6 \text{sinc}\left(\frac{3\omega}{\pi}\right)$
- (c) $\frac{1}{6} \text{sinc}\left(\frac{3\omega}{\pi}\right)$
- (d) $6 \text{sinc}\left(\frac{\omega}{3\pi}\right)$

33. $u(t) = p_{0.5}(t)$ (a pulse of width 0.5 s). At what smallest positive frequency ω does $U(j\omega)$ first cross zero?

Answer: _____

34. $u(t) = p_4(t)$. Evaluate $U(j\pi/4)$.

Answer: _____

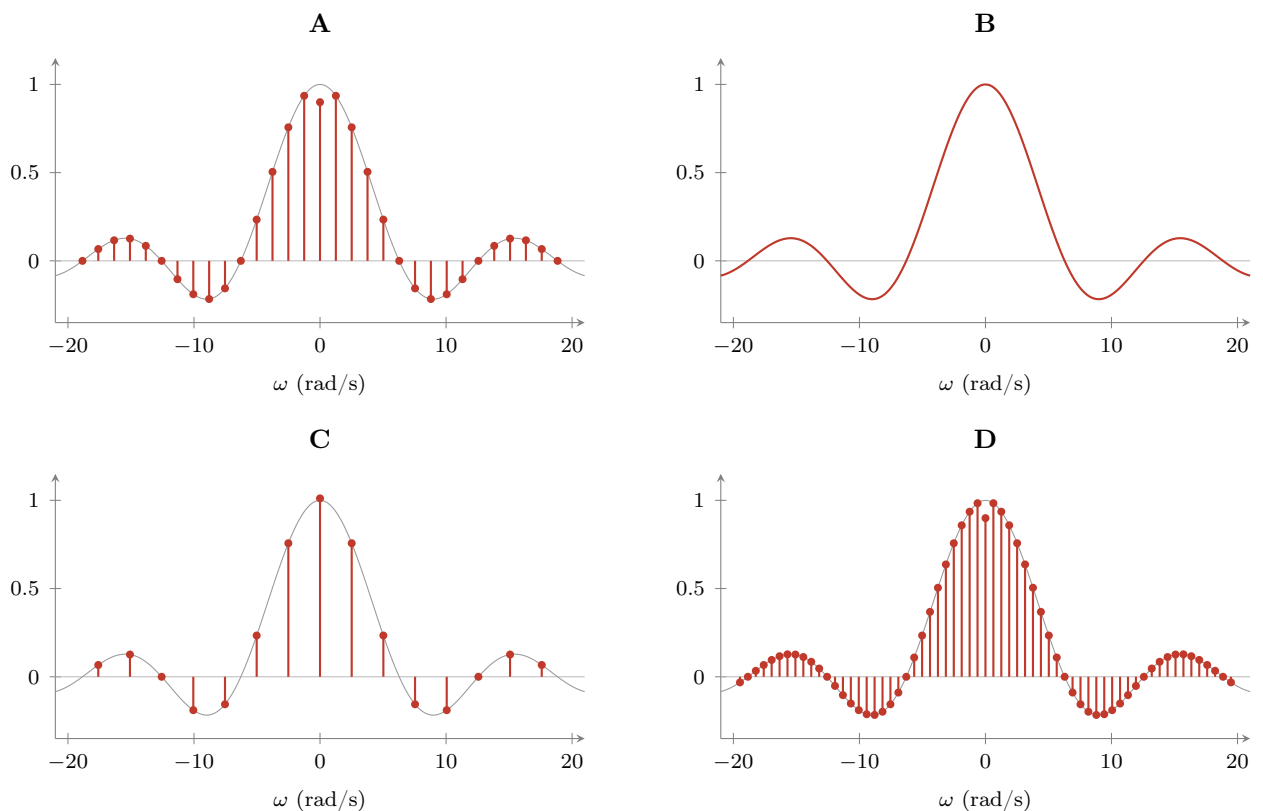
35. A rectangular pulse train has amplitude $A = 1$ and pulse width $\tau = 1$ s, and repeats with period T :

$$u_T(t) = \sum_{m \in \mathbb{Z}} p_1(t - mT).$$

Four signals are built this way, differing only in the period:

- (i) $T = 2.5$ s
- (ii) $T = 5$ s
- (iii) $T = 10$ s
- (iv) the single pulse $p_1(t)$, which is not periodic

Each plot below shows the spectrum of one of these signals against ω : the lines Tc_k at $\omega = k\omega_0$ for the three periodic signals, and $U(j\omega)$ for the single pulse. The grey curve on each plot is $\text{sinc}(\omega/2\pi)$. Match each of the four signals to its plot.



Write the matching letter on each line. Half a mark for each correct match.

- (i) _____ (ii) _____ (iii) _____ (iv) _____

36. $u(t) = e^{-3t}H(t)$. Evaluate $|U(j4)|$.

Answer: _____

37. $u(t) = e^{-a|t|}$ has Fourier transform $U(j\omega) = \frac{2a}{a^2 + \omega^2}$. For $a = 5000 \text{ s}^{-1}$, find $|U(j\omega)|$ at $\omega = a = 5000 \text{ rad/s}$.

Answer: _____

38. A system has $G(j\omega) = \frac{1}{3 + j\omega}$. Find the phase of $G(j4)$ in degrees.

Answer: _____

Properties of the Fourier transform

39. $u(t) = p_2(t)$. Compare $U(j\omega)$ with the transform of the delayed pulse $u(t - 5)$. Which statement is correct?
- (a) $|U(j\omega)|$ gains an extra term -5ω ; the phase is unchanged
 - (b) Neither the magnitude nor the phase spectrum changes
 - (c) $|U(j\omega)|$ is unchanged; the phase gains an extra term -5ω
 - (d) The whole spectrum is multiplied by $e^{-j5\omega}$, changing both magnitude and phase
40. $u(t)$ has spectrum $U(j\omega)$. The signal is compressed in time by a factor of 3, giving $u(3t)$. How do the **width** and the **peak height** of the spectrum change?
- (a) The spectrum compresses by a factor of 3 in ω and its height scales by 3
 - (b) Both the width and the height scale by 3
 - (c) The spectrum expands by a factor of 3 in ω and its height scales by $1/3$
 - (d) The spectrum expands by a factor of 3 in ω and its height is unchanged
41. Let $y(t) = p_4(2t - 4)$. Using the time-shift and time-scaling properties together with the pair $p_\tau(t) \rightsquigarrow \tau \operatorname{sinc}(\omega\tau/2\pi)$, evaluate $Y(j\omega)$ at $\omega = \pi/2$.

Answer: _____

42. The width- τ triangle $1 - |t|/\tau$ ($|t| < \tau$) has transform $\tau \operatorname{sinc}^2(\omega\tau/2\pi)$. Applying the duality property to this pair, what kind of function of ω is the Fourier transform of a **time-domain** signal shaped like $\operatorname{sinc}^2(t/\tau)$?
- (a) A rectangular pulse in ω
 - (b) A triangular function of ω
 - (c) Another sinc^2 -shaped function of ω
 - (d) A train of impulses in ω
43. What is the Fourier transform of $u(t) = \frac{2a}{t^2 + a^2}$, where $a > 0$?
- (a) $2\pi e^{-a|\omega|}$
 - (b) $e^{-a|\omega|}$
 - (c) $\frac{2a}{a^2 + \omega^2}$
 - (d) $2\pi e^{-a\omega}$ (for all ω)

44. Given that $u(t) \rightsquigarrow U(j\omega)$, which of the following correctly states the duality property of the Fourier transform?

- (a) $U(jt) \rightsquigarrow 2\pi u(\omega)$
- (b) $U(jt) \rightsquigarrow 2\pi u(-\omega)$
- (c) $U(jt) \rightsquigarrow u(-\omega)$
- (d) $U(-jt) \rightsquigarrow 2\pi u(-\omega)$
45. Which Fourier-transform property converts **convolution in time** into a **product in frequency**?
- (a) The differentiation property: $\frac{d}{dt}u(t) \rightsquigarrow j\omega U(j\omega)$
- (b) The convolution property: $h(t) * u(t) \rightsquigarrow G(j\omega)U(j\omega)$
- (c) The duality property: $U(jt) \rightsquigarrow 2\pi u(-\omega)$
- (d) The multiplication property: $u_1(t)u_2(t) \rightsquigarrow \frac{1}{2\pi}(U_1 * U_2)(j\omega)$
46. Which property converts a **product of two time signals** into a (scaled) **convolution of their transforms**?
- (a) The convolution property: $h(t) * u(t) \rightsquigarrow G(j\omega)U(j\omega)$
- (b) The integration property: $\int_{-\infty}^t u(\varsigma) d\varsigma \rightsquigarrow U(j\omega)/j\omega + \pi U(0)\delta(\omega)$
- (c) The multiplication property: $u_1(t)u_2(t) \rightsquigarrow \frac{1}{2\pi}(U_1 * U_2)(j\omega)$
- (d) The scaling property: $u(at) \rightsquigarrow \frac{1}{|a|}U(j\omega/a)$
47. What is the Fourier transform of $\frac{d}{dt} \cos(\omega_0 t)$?
- (a) $j\omega_0 \pi [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$
- (b) $j\pi\omega_0 [\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$
- (c) $-j\pi [\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$
- (d) $\pi [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$
48. What is the Fourier transform of $\frac{d}{dt} [H(t)e^{-3t}]$?
- (a) $\frac{-3}{3 + j\omega}$
- (b) 1
- (c) $\frac{j\omega}{(3 + j\omega)^2}$
- (d) $\frac{j\omega}{3 + j\omega}$
49. A signal $u(t)$ has Fourier transform $U(j\omega)$, and it is known that $U(j5) = 2 - 3j$. Let $y(t) = \frac{du}{dt}$. Evaluate $Y(j5)$.

Answer: _____

The transform of a periodic signal

50. $u(t)$ is periodic with period T , fundamental frequency $\omega_0 = 2\pi/T$, and Fourier series coefficients c_k . What does $U(j\omega)$ look like?

(a) $U(j\omega) = 2\pi \sum_{k=-\infty}^{\infty} c_k \delta(\omega - k\omega_0)$

(b) $U(j\omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_0)$

(c) $U(j\omega) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t}$

(d) $U(j\omega) = \sum_{k=-\infty}^{\infty} c_k \delta(\omega - k\omega_0)$

51. $u(t) = 5 + 4\cos(3t) - 2\sin(3t)$ is periodic with fundamental frequency $\omega_0 = 3$ rad/s. Writing $U(j\omega)$ as a sum of weighted impulses, find the complex weight of the impulse located at $\omega = 3$.

Answer: _____

52. A periodic, real-valued signal $u(t)$ has Fourier transform

$$U(j\omega) = 2\pi [3\delta(\omega) + (1 - 2j)\delta(\omega - 5) + (1 + 2j)\delta(\omega + 5)].$$

Find $u(0)$.

Answer: _____

Parseval's theorem and signal energy

53. Which of the following correctly states Parseval's theorem for the Fourier transform?

(a) $\int_{-\infty}^{\infty} |u(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |U(j\omega)|^2 d\omega$

(b) $\int_{-\infty}^{\infty} |u(t)|^2 dt = \int_{-\infty}^{\infty} |U(j\omega)|^2 d\omega$

(c) $\int_{-\infty}^{\infty} u(t) dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} U(j\omega) d\omega$

(d) $\int_{-\infty}^{\infty} |u(t)|^2 dt = \frac{1}{\pi} \int_{-\infty}^{\infty} |U(j\omega)|^2 d\omega$

54. Find the energy of $u(t) = \frac{\sin(3t)}{\pi t}$.

Answer: _____

3. Frequency response (week 6)

Evaluating and using $G(j\omega)$

55. A system has $G(j\omega) = \frac{j\omega + 2}{(j\omega)^2 + 5(j\omega) + 4}$. Find the steady-state response to the input $u(t) = 5\cos(2t + 30^\circ)$.

Answer: _____

56. A system has $G(j\omega) = \frac{j\omega + 0.1}{j\omega + 5}$. Its steady-state response to $u(t) = \cos(10t - 50^\circ)$ is $y(t) =$

- (a) $0.372 \cos(10t - 24.0^\circ)$
- (b) $0.894 \cos(10t - 24.0^\circ)$
- (c) $1.00 \cos(10t - 50.0^\circ)$
- (d) $0.894 \cos(10t - 76.0^\circ)$

57. A vehicle's height h and road height r are related by $m\ddot{h} = -k(h - r) - b(\dot{h} - \dot{r})$ with $m = 1$, $k = 2$, $b = 3$ (SI units). For a sinusoidal road profile $r(t) = \cos(2t)$, the vehicle height reaches steady state. Find its RMS value.

Answer: _____

58. A system has $G(j\omega) = \frac{2(j\omega)^2 + 4}{(j\omega)^2 + 4(j\omega) + 4}$. Find $|G(j2)|$ and $\arg G(j2)$.

Answer: _____

59. A system has $G(j\omega) = \frac{300}{j\omega + 4}$. Find the phase of $G(j4)$ in degrees.

Answer: _____

60. A river's height $y(t)$ (in feet) responds to rainfall rate $u(t)$ (in/hr) via $G(j\omega) = \frac{10}{(9j\omega + 1)(16j\omega + 1)}$, with time in hours. Find $|G(j\omega)|$ at $\omega = 1/12$ rad/hr.

Answer: _____

61. A system obeys $\ddot{y} + 3\dot{y} + 2y = u$, so $G(j\omega) = \frac{1}{(j\omega + 1)(j\omega + 2)}$. Find the steady-state output for the constant input $u(t) = 1$.

Answer: _____

62. A full-wave rectified sine $u(t) = |\sin t|$ (period $T = \pi$, $\omega_0 = 2$) drives a first-order lowpass filter with $G(j\omega) = \frac{1}{1 + 3j\omega}$. Find the DC component of the output $y(t)$.

Answer: _____

63. A system has impulse response $h(t) = 3e^{-6t}H(t)$, so $G(j\omega) = \frac{3}{j\omega + 6}$. What is the steady-state output when the input is the constant signal $u(t) = 4$ for all t ?

- (a) $y = 0.500$
- (b) $y = 8.00$
- (c) $y = 12.0$
- (d) $y = 2.00$

Bode diagrams

64. The straight-line (asymptotic) Bode magnitude plot of a first-order system is flat at 12 dB for $\omega < 200$ rad/s and falls at -20 dB/decade for $\omega > 200$ rad/s, with its corner at $\omega = 200$ rad/s. Reading off that plot, what is $20 \log_{10} |G(j2000)|$?

Answer: _____

65. The straight-line (asymptotic) Bode phase plot of a system with a single real pole at $\omega_c = 1000$ rad/s is 0° for $\omega \leq \omega_c/10$, a straight segment of slope -45° per decade from $\omega_c/10$ to $10\omega_c$, and -90° for $\omega \geq 10\omega_c$. Reading off that plot, what is the phase at $\omega = 10,000$ rad/s?

- (a) -90°
- (b) -84.29°
- (c) 0°
- (d) -45°

66. A system has impulse response $h(t) = e^{-at}H(t)$ with $a = 3$, so $G(j\omega) = \frac{1}{a + j\omega}$. At what frequency ω (in rad/s, $\omega > 0$) does $|G(j\omega)|$ fall to $1/\sqrt{2}$ of its value $|G(j0)|$ at zero frequency?

Answer: _____

67. Four systems:

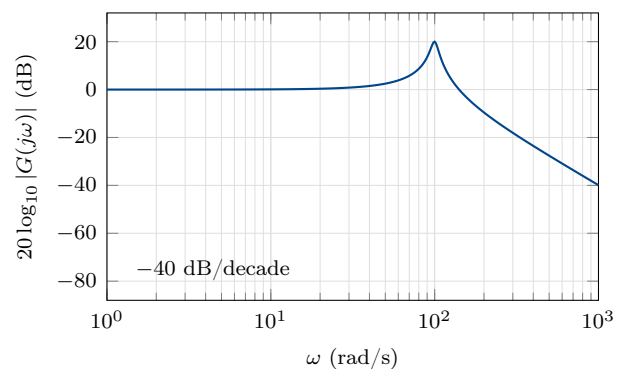
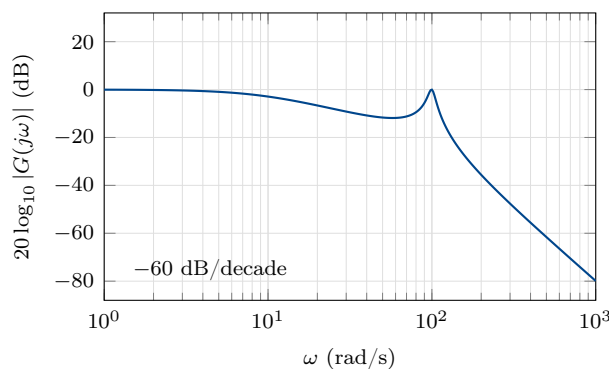
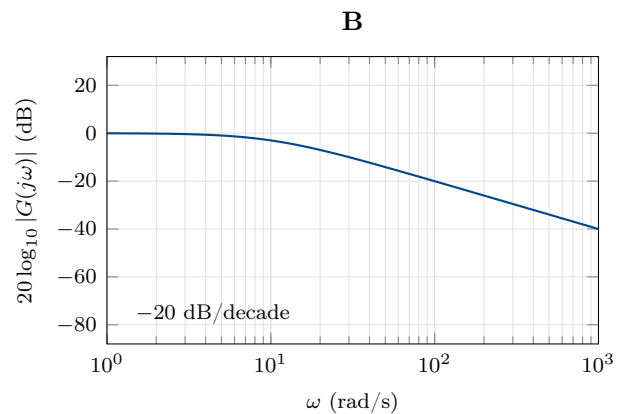
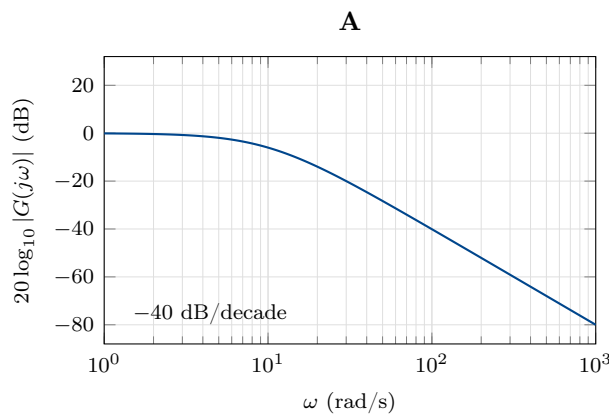
$$(i) \quad G(j\omega) = \frac{1}{1 + j\omega/10}$$

$$(ii) \quad G(j\omega) = \frac{10^4}{(j\omega)^2 + 10(j\omega) + 10^4}$$

$$(iii) \quad G(j\omega) = \frac{100}{(j\omega)^2 + 20(j\omega) + 100}$$

$$(iv) \quad G(j\omega) = \frac{1}{1 + j\omega/10} \cdot \frac{10^4}{(j\omega)^2 + 10(j\omega) + 10^4}$$

Their Bode magnitude diagrams, in some order, are below. Match each system to its diagram.



Write the matching letter on each line. Half a mark for each correct match.

(i) _____

(ii) _____

(iii) _____

(iv) _____

Moving between the impulse response and $G(j\omega)$

68. A system has $G(j\omega) = \frac{4}{j\omega + 2}$. Find its impulse response $h(t)$.

Answer: _____

69. A first-order system with impulse response $h_1(t) = 2e^{-5t}H(t)$ is connected in cascade with an ideal delay of 3 seconds (impulse response $\delta(t-3)$). Find the impulse response $h(t)$ of the overall cascade system.

Answer: _____

70. A system has $G(j\omega) = \frac{3}{j\omega + 1} + \frac{2}{j\omega + 4}$. What is its impulse response $h(t)$?

- (a) $h(t) = 5e^{-5t}H(t)$
- (b) $h(t) = 3e^{-4t}H(t) + 2e^{-t}H(t)$
- (c) $h(t) = 3e^{-t}H(t) - 2e^{-4t}H(t)$
- (d) $h(t) = 3e^{-t}H(t) + 2e^{-4t}H(t)$

71. A system has impulse response

$$h(t) = 4e^{-3t} \sin(5t) H(t).$$

Its transfer function $G(j\omega)$ is:

- (a) $\frac{4}{(3 + j\omega)^2 + 25}$
- (b) $\frac{20}{(5 + j\omega)^2 + 9}$
- (c) $\frac{4(3 + j\omega)}{(3 + j\omega)^2 + 25}$
- (d) $\frac{20}{(3 + j\omega)^2 + 25}$

4. Second order systems and stability (weeks 6-7)

Damping and resonance in second order systems

72. Which damping ratio makes a second-order system $G(j\omega) = \frac{K\omega_n^2}{(j\omega)^2 + 2\zeta\omega_n(j\omega) + \omega_n^2}$ critically damped?

- (a) $\zeta = 1.5$
- (b) $\zeta = 0$
- (c) $\zeta = 1.0$
- (d) $\zeta = 0.4$

73. A second-order system in the standard form $(j\omega)^2 + 2\zeta\omega_n(j\omega) + \omega_n^2$ has its two poles at $-10^4 \pm j10^4$ rad/s. What is ζ ?

- (a) 0.5
- (b) 1
- (c) 0.707
- (d) 1.414

74. A second-order system has $\zeta = 0.8$. Does $|G(j\omega)|$ have a resonant peak at a nonzero frequency?

- (a) No, because $\zeta = 0.8 > 1/\sqrt{2} \approx 0.7071$
- (b) It cannot be determined without knowing K
- (c) Yes, at $\omega_r = \omega_n \sqrt{1 - 2(0.8)^2}$

(d) Yes, because any $\zeta < 1$ (underdamped) gives a resonant peak

75. A series RLC circuit is driven by $u(t)$, with the output $y(t)$ taken across the capacitor, so $LC\ddot{y} + RC\dot{y} + y = u$. Here $R = 200\ \Omega$, $L = 0.5\ \text{H}$ and $C = 2\ \mu\text{F}$. Does $|G(j\omega)|$ have a resonant peak? If so, find ω_r and the peak value $|G(j\omega_r)|$ in dB.

Answer: _____

76. A designer wants a sharp resonant peak near $\omega_0 = 5\ \text{rad/s}$, using a pair of complex conjugate poles at $-\alpha \pm j5$. Which choice of α gives the sharpest, highest resonant peak?

- (a) $\alpha = 2$
- (b) $\alpha = 0.5$
- (c) $\alpha = 0.05$
- (d) $\alpha = 5$

Stability from the poles

77. An LTI system has $G(j\omega) = \frac{(j\omega)^2 + j\omega + 1}{(j\omega + 2)((j\omega)^2 + 4)}$. Classify the system.

- (a) Marginally stable: no pole in the right half plane, and the poles on the imaginary axis are distinct
- (b) Unstable: no pole in the right half plane, but a repeated pole on the imaginary axis
- (c) Unstable: at least one pole satisfies $\text{Re } p > 0$
- (d) Stable: every pole satisfies $\text{Re } p < 0$

78. An LTI system has $G(j\omega) = \frac{(j\omega)^2 + 2(j\omega) + 8}{(j\omega + 1)((j\omega)^2 + 4)^2}$. Classify the system.

- (a) Unstable: at least one pole satisfies $\text{Re } p > 0$
- (b) Unstable: no pole in the right half plane, but a repeated pole on the imaginary axis
- (c) Marginally stable: no pole in the right half plane, and the poles on the imaginary axis are distinct
- (d) Stable: every pole satisfies $\text{Re } p < 0$

79. Four LTI systems are given by their transfer functions. Which one is BIBO stable?

(a) $G(j\omega) = \frac{j\omega + 7}{(j\omega + 1)((j\omega)^2 - 4(j\omega) + 9)}$

(b) $G(j\omega) = \frac{1}{j\omega - 3}$

(c) $G(j\omega) = \frac{2(j\omega) + 3}{(j\omega + 1)(j\omega + 2)}$

(d) $G(j\omega) = \frac{3}{j\omega(j\omega + 2)}$

80. A first-order system obeys $\tau\dot{y} = -y + \beta u$ with $\tau = -4\ \text{s}$. Classify its stability.

- (a) Marginally stable
- (b) Stable and BIBO stable
- (c) Cannot be determined without knowing β
- (d) Unstable

81. For the second-order system $G(j\omega) = \frac{K\omega_n^2}{(j\omega)^2 + 2\zeta\omega_n(j\omega) + \omega_n^2}$ with $\omega_n = 5$ rad/s fixed, for which value of ζ is the unforced response *not* bounded as $t \rightarrow \infty$?

- (a) $\zeta = 1$
- (b) $\zeta = -0.1$
- (c) $\zeta = 0.3$
- (d) $\zeta = 2$

82. For $ay'' + by' + cy = 0$ with $a, b, c > 0$, which is always true of the solution $y(t)$ as $t \rightarrow \infty$?

- (a) $y(t)$ can grow without bound for some positive choices of a, b, c
- (b) $y(t)$ stays bounded but does not necessarily decay to zero
- (c) $y(t) \rightarrow 0$, regardless of the relative sizes of a, b, c
- (d) $y(t) \rightarrow 0$ only if $b^2 \geq 4ac$ (overdamped or critically damped)

BIBO stability from the impulse response

83. A system has impulse response $h(t) = 3e^{-2t}H(t)$. Compute $\int_{-\infty}^{\infty} |h(t)| dt$ and state whether the system is BIBO stable.

Answer: _____

84. A system has impulse response $h(t) = H(t)$ (an ideal integrator). Classify its pole, its stability, and its BIBO stability.

- (a) Pole at $j\omega = 0$; marginally stable; BIBO stable
- (b) Pole at $j\omega = -1$; stable; BIBO stable
- (c) Pole at $j\omega = 0$; marginally stable; BIBO unstable
- (d) No finite poles; the output depends only on the present value of the input

85. Two systems have impulse responses $h_A(t) = 300e^{-4t}H(t)$ and $h_B(t) = 300e^{4t}H(t)$. Which system is BIBO stable and has a frequency response $G(j\omega)$ that exists (that is, its Fourier transform converges)?

- (a) neither
- (b) A only
- (c) both A and B
- (d) B only